

COMPUTER SCIENCE TRIPOS Part IB – 2021 – Paper 6

3 Complexity Theory (mpf23)

CNF, SAT

- (a) Define the set of Boolean expressions 2CNF and the language 2SAT over them. [2 marks]

Answer: A Boolean expression ϕ is said to be in 2CNF if it is in CNF, say $\phi \equiv C_1 \wedge \dots \wedge C_n$, and each clause C_i is the disjunction of 2 literals. We define 2SAT to be the set of those expressions in 2CNF that are satisfiable.

2SAT

- (b) For a Boolean expression ϕ in 2CNF, let $G(\phi)$ be the directed graph with *vertices* the variables of ϕ and their negation, and with *edges* (a, b) if, and only if, there is a clause $(\neg a \vee b)$ or $(b \vee \neg a)$ in ϕ . Note that an edge (a, b) is in $G(\phi)$ if, and only if, so is the edge $(\neg b, \neg a)$.

Prove that a Boolean expression ϕ in 2CNF is unsatisfiable if, and only if, there is a variable x in ϕ such that there are paths from x to $\neg x$ and from $\neg x$ to x in $G(\phi)$. [*Hint:* Recall that the proposition $(\neg P \vee Q)$ is equivalently the implication $(P \rightarrow Q)$.] [12 marks]

Answer: Suppose that such paths exist, and still ϕ can be satisfied by a truth assignment T . Suppose that $T(x) = \text{true}$ (a similar argument works when $T(x) = \text{false}$). Since there is a path from x to $\neg x$, and $T(x) = \text{true}$ while $T(\neg x) = \text{false}$, there must be an edge (a, b) along this path such that $T(a) = \text{true}$ and $T(b) = \text{false}$. However, since (a, b) is an edge of $G(\phi)$, it follows that $(\neg a \vee b)$ or $(b \vee \neg a)$ is a clause of ϕ . This clause is not satisfied by T , a contradiction.

Conversely, suppose that there is no variable with such paths in $G(\phi)$; that is, for every variable x of ϕ there is no path from x to $\neg x$ or no path from $\neg x$ to x in $G(\phi)$. We are going to construct a satisfying truth assignment, that is, a truth assignment such that no edge of $G(\phi)$ goes from **true** to **false**. We repeat the following step: We pick a node a whose truth value has not yet been defined, and such that there is no path from a to $\neg a$. We consider all nodes reachable from a in $G(\phi)$, and assign them the value **true**. We also assign **false** to the negations of these nodes (the negations correspond to all these nodes from which $\neg a$ is reachable). This step is well-defined because, if there were paths from a to both b and $\neg b$, then there would be paths to $\neg a$ from both of these (by the symmetry of $G(\phi)$), and therefore a path from a to $\neg a$, a contradiction of our hypothesis. Furthermore, if there were a path from a to a node already assigned **false** in a previous step, then a is a predecessor of that node, and was also assigned **false** at that step.

We repeat this step until all nodes have a truth assignment. Since we assumed that there are no paths from any x to $\neg x$ and back, all nodes will be assigned a truth value. And since the steps are such that, whenever a node is assigned **true** all of its successors are also assigned **true**, and the opposite for **false**, there can be no edge from **true** to **false**. The truth assignment satisfies ϕ .

Reachability, NL,
P, NP

- (c) Argue as to whether or not 2SAT is in NL, in P, and in NP. Your answer may use the fact that NL is closed under complementation. [6 marks]

Answer: 2SAT is in NL and hence also in P and in NP.

Since NL is closed under complementation, it suffices to show that we can recognize unsatisfiable 2CNF expressions in NL. In nondeterministic logarithmic space we can test the

— *Solution notes* —

condition of the characterisation in part (b) by guessing a variable x , and paths from x to $\neg x$ and back.
